

PRACTICAL NO. 8

Aim: Set up and configure an intrusion detection system (IDS) to monitor network traffic and detect potential security breaches or malicious activities.

OUTPUT: UBUNTU

```

it-24@IT-24:~$ sudo apt update
[sudo: authenticate] Password:
Hit:1 http://archive.ubuntu.com/ubuntu resolute InRelease
Hit:2 http://security.ubuntu.com/ubuntu resolute-security InRelease
Hit:3 http://archive.ubuntu.com/ubuntu resolute-updates InRelease
Hit:4 http://archive.ubuntu.com/ubuntu resolute-backports InRelease
2 packages can be upgraded. Run 'apt list --upgradable' to see them.
it-24@IT-24:~$ sudo apt upgrade -y
Not upgrading yet due to phasing:
python3-software-properties software-properties-common

Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 2
it-24@IT-24:~$ sudo apt install snort -y
Error: Unable to locate package snort
it-24@IT-24:~$

28°C
Mostly cloudy
09:57
24-08-2026

it-24@IT-24:~$ sudo apt install suricata -y
Installing:
  suricata

Installing dependencies:
  isa-support          librtt-bus-pci26      librtt-net26
  libevent-2.1-7t64   librtt-bus-vdev26     librtt-pci26
  libevent-pthreads-2.1-7t64  librtt-eal26         librtt-rcu26
  libfdt1             librtt-ethdev26       librtt-ring26
  libhiredis1.1.0     librtt-hash26         librtt-sched26
  libhyperscan5       librtt-ip-frag26      librtt-telemetry26
  libmaximindb0       librtt-kvargs26       libxdpi
  libnet9             librtt-log26          sse3-support
  libnetfilter-log1   librtt-mbuf26         sse4.2-support
  libnetfilter-queue1 librtt-mempool26      suricata-update
  libnfnetlink0       librtt-meter26
  librtt-argparse26   librtt-net-bond26

Suggested packages:
  mmdns-bin libtcmalloc-minimal4

Summary:
  Upgrading: 0, Installing: 35, Removing: 0, Not Upgrading: 2
  Download size: 8942 kB
  Space needed: 38.6 MB / 1022 GB available

Get:1 http://archive.ubuntu.com/ubuntu resolute/main amd64 isa-support amd64 27ubuntu2 [16.9 kB]
Get:2 http://archive.ubuntu.com/ubuntu resolute/universe amd64 sse3-support amd64 27ubuntu2 [3702 B]
Get:3 http://archive.ubuntu.com/ubuntu resolute/main amd64 sse4.2-support amd64 27ubuntu2 [3670 B]
Get:4 http://archive.ubuntu.com/ubuntu resolute/main amd64 libmaximindb0 amd64 1.12.2-1build2 [18.9 kB]
Get:5 http://archive.ubuntu.com/ubuntu resolute/main amd64 libnfnetlink0 amd64 1.0.2-3build1 [15.1 kB]
Get:6 http://archive.ubuntu.com/ubuntu resolute/main amd64 libevent-2.1-7t64 amd64 2.1.12-stable-10build2 [145 kB]
Get:7 http://archive.ubuntu.com/ubuntu resolute/main amd64 libevent-pthreads-2.1-7t64 amd64 2.1.12-stable-10build2 [8146 B]
Get:8 http://archive.ubuntu.com/ubuntu resolute/universe amd64 libhiredis1.1.0 amd64 1.2.0-6ubuntu4 [41.9 kB]
Get:9 http://archive.ubuntu.com/ubuntu resolute/universe amd64 libhyperscan5 amd64 5.4.2-4 [3003 kB]
Get:10 http://archive.ubuntu.com/ubuntu resolute/main amd64 libnet9 amd64 1.3+dfsg-3 [43.8 kB]
Get:11 http://archive.ubuntu.com/ubuntu resolute/universe amd64 libnetfilter-log1 amd64 1.0.2-4build2 [13.9 kB]
Get:12 http://archive.ubuntu.com/ubuntu resolute/universe amd64 libnetfilter-queue1 amd64 1.0.5-4build2 [15.5 kB]
Get:13 http://archive.ubuntu.com/ubuntu resolute/main amd64 libfdt1 amd64 1.7.2-2ubuntu1 [21.3 kB]
Get:14 http://archive.ubuntu.com/ubuntu resolute/main amd64 librtt-log26 amd64 25.11-2 [25.4 kB]
Get:15 http://archive.ubuntu.com/ubuntu resolute/main amd64 librtt-argparse26 amd64 25.11-2 [11.5 kB]
Get:16 http://archive.ubuntu.com/ubuntu resolute/main amd64 librtt-kvargs26 amd64 25.11-2 [7556 B]

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Processing triggers for man-db (2.13.1-1build1) ...
Processing triggers for libc-bin (2.43-2ubuntu2.3) ...
it-24@IT-24:~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet 10.255.255.254/32 brd 10.255.255.254 scope global lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host proto kernel_lo
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1450 qdisc mq state UP group default qlen 1000
    link/ether 00:15:5d:a1:e9:8c brd ff:ff:ff:ff:ff:ff
    altname enx00155da1e98c
    inet 172.25.20.202/20 brd 172.25.31.255 scope global eth0
        valid_lft forever preferred_lft forever
    inet6 fe80::215:5dff:fe:a1:e98c/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever
it-24@IT-24:~$ sudo nano /etc/suricata/rules/local.rules
it-24@IT-24:~$ sudo nano /etc/suricata/suricata.yaml
it-24@IT-24:~$ sudo systemctl stop suricata
it-24@IT-24:~$ sudo suricata -c /etc/suricata/suricata.yaml -i eth0
i: suricata: This is Suricata version 0.0.3 RELEASE running in SYSTEM mode
W: detect: No rule files match the pattern /var/lib/suricata/rules/suricata.rules
W: detect: No rule files match the pattern /var/lib/suricata/rules/local.rules
W: detect: 2 rule files specified, but no rules were loaded!
i: mpm-hs: Rule group caching - loaded: 0 newly cached: 0 total cacheable: 0
i: threads: Threads created -> W: 12 FM: 1 FR: 1 Engine started.
^Ci: suricata: Signal Received. Stopping engine.
i: device: eth0: packets: 0, drops: 0 (0.00%), invalid chksum: 0
it-24@IT-24:~$ sudo nano /var/lib/suricata/rules/local.rules
it-24@IT-24:~$ sudo suricata -c /etc/suricata/suricata.yaml -i eth0
i: suricata: This is Suricata version 0.0.3 RELEASE running in SYSTEM mode
W: detect: No rule files match the pattern /var/lib/suricata/rules/suricata.rules
i: mpm-hs: Rule group caching - loaded: 0 newly cached: 0 total cacheable: 0
i: threads: Threads created -> W: 12 FM: 1 FR: 1 Engine started.
^Ci: suricata: Signal Received. Stopping engine.
i: device: eth0: packets: 6, drops: 0 (0.00%), invalid chksum: 0
it-24@IT-24:~$
it-24@IT-24:~$ sudo touch /var/lib/suricata/rules/suricata.rules
it-24@IT-24:~$ sudo suricata -c /etc/suricata/suricata.yaml -i eth0 -S /var/lib/suricata/rules/local.rules

it-24@IT-24:~$ sudo suricata -c /etc/suricata/suricata.yaml -i eth0 -S /var/lib/suricata/rules/local.rules -k none -A console
suricata: invalid option -- 'A'
Suricata 0.0.3 ("undefined")
USAGE: suricata [OPTIONS] [BPF FILTER]

General:
  -v                : be more verbose (use multiple times to increase verbosity)
  -c <path>         : path to configuration file
  -l <dir>          : default log directory
  --include <path>  : additional configuration file
  --set name=value   : set a configuration value
  --pidfile <file>  : write pid to this file
  -T               : test configuration file (use with -c)
  --init-errors=fatal : enable fatal failure on signature init error
  -D              : run as daemon
  --user <user>     : run suricata as this user after init
  --group <group>   : run suricata as this group after init
  --unix-socket=<file> : use unix socket to control suricata work
  --runmode <runmode-id> : specific runmode modification the engine should run. The argument
                        : supplied should be the id for the runmode obtained by running
                        : --list-runmodes

Capture and IPS:
  -F <bpf filter file> : bpf filter file
  -k [all|none]        : force checksum check (all) or disabled it (none)
  -i <dev or ip>       : run in pcap live mode
  --pcap=<dev>         : run in pcap mode, no value select interfaces from suricata.yaml
  --pcap-buffer-size   : size of the pcap buffer value from 0 - 2147483647
  -q <qid[:qid]>        : run in inline nqueue mode (use colon to specify a range of queues)
  --af-packet[=<dev>]  : run in af-packet mode, no value select interfaces from suricata.yaml
  --af-xdp[=<dev>]    : run in af-xdp mode, no value select interfaces from suricata.yaml
  --dpdk              : run in dpdk mode, uses interfaces from suricata.yaml
  --reject-dev <dev>  : send reject packets from this interface

Capture Files:
  -r <path>          : run in pcap file/offline mode
  --pcap-file-continuous : when running in pcap mode with a directory, continue checking directory for pcaps until interrupted
  --pcap-file-delete    : when running in replay mode (-r with directory or file), will delete pcap files that have been processed when done
  --pcap-file-recursive : will descend into subdirectories when running in replay mode (-r)
  --pcap-buffer-size    : set read buffer size (setvbuf)
  --erf-in <path>      : process an ERF file

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it-24@IT-24:~$ sudo tcpdump -i any -c 5 icmp -w /tmp/test.pcap & ping -c 2 127.0.0.1
[1] 24505
PING 127.0.0.1 (127.0.0.1) 56(84) bytes of data.
64 bytes from 127.0.0.1: icmp_seq=1 ttl=64 time=0.073 ms
sudo: 'tcpdump': command not found
64 bytes from 127.0.0.1: icmp_seq=2 ttl=64 time=0.032 ms

--- 127.0.0.1 ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 1019ms
rtt min/avg/max/mdev = 0.032/0.052/0.073/0.020 ms
[1]+  Exit 1                  sudo tcpdump -i any -c 5 icmp -w /tmp/test.pcap
it-24@IT-24:~$ sudo suricata -c /etc/suricata/suricata.yaml -S /var/lib/suricata/rules/local.rules -k none -l /tmp/suricata_test -r /tmp/test.pcap
Error: suricata: pcap file '/tmp/test.pcap': No such file or directory
it-24@IT-24:~$ cat /tmp/suricata_test/fast.log
it-24@IT-24:~$ sudo apt update && sudo apt install tcpdump -y
Hit:1 http://archive.ubuntu.com/ubuntu resolute InRelease
Hit:2 http://security.ubuntu.com/ubuntu resolute-security InRelease
Hit:3 http://archive.ubuntu.com/ubuntu resolute-updates InRelease
Hit:4 http://archive.ubuntu.com/ubuntu resolute-backports InRelease
2 packages can be upgraded. Run 'apt list --upgradable' to see them.
Installing:
  tcpdump
Summary:
  Upgrading: 0, Installing: 1, Removing: 0, Not Upgrading: 2
  Download size: 477 kB
  Space needed: 1357 kB / 1022 GB available
Get:1 http://archive.ubuntu.com/ubuntu resolute/main amd64 tcpdump amd64 4.99.6-1 [477 kB]
Fetched 477 kB in 0s (1850 kB/s)
Selecting previously unselected package tcpdump.
(Reading database ... 50701 files and directories currently installed.)
Preparing to unpack .../tcpdump_4.99.6-1_amd64.deb ...
Unpacking tcpdump (4.99.6-1) ...
Setting up tcpdump (4.99.6-1) ...
Creating group 'tcpdump' with GID 986.
Creating user 'tcpdump' (tcpdump) with UID 986 and GID 986.
Processing triggers for man-db (2.13.1-1build1) ...
it-24@IT-24:~$ sudo tcpdump -i any -c 5 icmp -w /tmp/test.pcap & ping -c 2 127.0.0.1
[1] 24732
PING 127.0.0.1 (127.0.0.1) 56(84) bytes of data.
64 bytes from 127.0.0.1: icmp_seq=1 ttl=64 time=2.74 ms

Creating user 'tcpdump' (tcpdump) with UID 986 and GID 986.
Processing triggers for man-db (2.13.1-1build1) ...
it-24@IT-24:~$ sudo tcpdump -i any -c 5 icmp -w /tmp/test.pcap & ping -c 2 127.0.0.1
[1] 24732
PING 127.0.0.1 (127.0.0.1) 56(84) bytes of data.
64 bytes from 127.0.0.1: icmp_seq=1 ttl=64 time=2.74 ms
tcpdump: WARNING: any: That device doesn't support promiscuous mode
(Promiscuous mode not supported on the "any" device)
tcpdump: listening on any, link-type LINUX_SLL2 (Linux cooked v2), snapshot length 262144 bytes
64 bytes from 127.0.0.1: icmp_seq=2 ttl=64 time=0.031 ms

--- 127.0.0.1 ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 1002ms
rtt min/avg/max/mdev = 0.031/1.387/2.744/1.356 ms
it-24@IT-24:~$ sudo suricata -c /etc/suricata/suricata.yaml -S /var/lib/suricata/rules/local.rules -k none -l /tmp/suricata_test -r /tmp/test.pcap
i: suricata: This is Suricata version 8.0.3 RELEASE running in USER mode
i: mpm-hs: Rule group caching - loaded: 0 newly cached: 0 total cacheable: 0
E: pcap: truncated dump file; tried to read 4 file header bytes, only got 0
W: pcap: Failed to init pcap file /tmp/test.pcap, skipping
E: pcap: pcap file reader thread failed to initialize
i: threads: Threads created -> RX: 1 W: 12 FM: 1 FR: 1 Engine started.
i: suricata: Signal Received. Stopping engine.
it-24@IT-24:~$ sudo cat /tmp/suricata_test/fast.log
it-24@IT-24:~$ sudo tcpdump -i lo -c 4 icmp -w /tmp/test.pcap & sleep 1 && ping -c 2 127.0.0.1
[2] 24818
tcpdump: /tmp/test.pcap: Permission denied
PING 127.0.0.1 (127.0.0.1) 56(84) bytes of data.
64 bytes from 127.0.0.1: icmp_seq=1 ttl=64 time=0.038 ms
64 bytes from 127.0.0.1: icmp_seq=2 ttl=64 time=0.046 ms

--- 127.0.0.1 ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 1029ms
rtt min/avg/max/mdev = 0.038/0.042/0.046/0.004 ms
[2]+  Exit 1                  sudo tcpdump -i lo -c 4 icmp -w /tmp/test.pcap
it-24@IT-24:~$ sudo tcpdump -i any -c 5 icmp -w /tmp/test.pcap
12 packets received by filter
0 packets dropped by kernel
^C
[1]+  Done                  sudo tcpdump -i any -c 5 icmp -w /tmp/test.pcap
it-24@IT-24:~$ sudo tcpdump -i lo -c 4 icmp -w ~/test.pcap
tcpdump: listening on lo, link-type EN10MB (Ethernet), snapshot length 262144 bytes
4 packets captured

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it-24@IT-24:~$ sudo tcpdump -i lo -c 4 icmp -w ~/test.pcap
tcpdump: listening on lo, link-type EN10MB (Ethernet), snapshot length 262144 bytes
4 packets captured
8 packets received by filter
0 packets dropped by kernel
it-24@IT-24:~$ sudo suricata -c /etc/suricata/suricata.yaml -S /var/lib/suricata/rules/local.rules -k none -l /tmp/suricata_test -r ~/test.pcap
i: suricata: This is Suricata version 8.0.3 RELEASE running in USER mode
i: mpm-hs: Rule group caching - loaded: 0 newly cached: 0 total cacheable: 0
i: threads: Threads created -> RX: 1 W: 12 FM: 1 FR: 1 Engine started.
i: suricata: Signal Received. Stopping engine.
i: pcap: read 1 file, 4 packets, 392 bytes
it-24@IT-24:~$ sudo cat /tmp/suricata_test/fast.log
it-24@IT-24:~$ sudo grep "event_type:alert" /tmp/suricata_test/eve.json
it-24@IT-24:~$ sudo suricata -c /etc/suricata/suricata.yaml -S /var/lib/suricata/rules/local.rules -k none --set outputs.1.fast.enabled=yes -l /tmp/suricata_test -r ~/test.pcap
i: suricata: This is Suricata version 8.0.3 RELEASE running in USER mode
i: mpm-hs: Rule group caching - loaded: 0 newly cached: 0 total cacheable: 0
i: threads: Threads created -> RX: 1 W: 12 FM: 1 FR: 1 Engine started.
i: suricata: Signal Received. Stopping engine.
i: pcap: read 1 file, 4 packets, 392 bytes
it-24@IT-24:~$ sudo cat /tmp/suricata_test/fast.log
it-24@IT-24:~$ sudo suricata -c /etc/suricata/suricata.yaml -S /var/lib/suricata/rules/local.rules -k none -l /tmp/suricata_test -r ~/test.pcap
i: suricata: This is Suricata version 8.0.3 RELEASE running in USER mode
E: detect-parse: Duplicate signature "alert icmp any any -> $HOME_NET any (msg:"ICMP Ping Detected"; sid:1000001; rev:1;)"
E: detect: error parsing signature "alert icmp any any -> $HOME_NET any (msg:"ICMP Ping Detected"; sid:1000001; rev:1;)" from file /var/lib/suricata/rules/local.rules at line 1
E: detect-parse: Duplicate signature "alert tcp any any -> $HOME_NET any (flags:S; msg:"Possible TCP SYN Scan Detected"; sid:1000002; rev:1;)"
E: detect: error parsing signature "alert tcp any any -> $HOME_NET any (flags:S; msg:"Possible TCP SYN Scan Detected"; sid:1000002; rev:1;)" from file /var/lib/suricata/rules/local.rules at line 2
i: mpm-hs: Rule group caching - loaded: 0 newly cached: 0 total cacheable: 0
i: threads: Threads created -> RX: 1 W: 12 FM: 1 FR: 1 Engine started.
i: suricata: Signal Received. Stopping engine.
i: pcap: read 1 file, 4 packets, 392 bytes
it-24@IT-24:~$ echo -e "alert icmp any any -> any any (msg:"ICMP Ping Detected"; sid:1000001; rev:1;) \n alert tcp any any -> any any (flags:S; msg:"Possible TCP SYN Scan Detected"; sid:1000002; rev:1;)" | sudo cat /var/lib/suricata/rules/local.rules
alert icmp any any -> any any (msg:"ICMP Ping Detected"; sid:1000001; rev:1;)
alert tcp any any -> any any (flags:S; msg:"Possible TCP SYN Scan Detected"; sid:1000002; rev:1;)
it-24@IT-24:~$ sudo suricata -c /etc/suricata/suricata.yaml -k none -l /tmp/suricata_test -r ~/test.pcap
i: suricata: This is Suricata version 8.0.3 RELEASE running in USER mode
i: mpm-hs: Rule group caching - loaded: 0 newly cached: 0 total cacheable: 0
i: threads: Threads created -> RX: 1 W: 12 FM: 1 FR: 1 Engine started.
i: suricata: Signal Received. Stopping engine.

```

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i: pcap: read 1 file, 4 packets, 392 bytes
it-24@IT-24:~$ sudo cat /tmp/suricata_test/fast.log
08/24/2026-04:51:03.625457 [**] [1:1000001:1] ICMP Ping Detected [**] [Classification: (null)] [Priority: 3] {ICMP} 127.0.0.1:8 -> 127.0.0.1:0
it-24@IT-24:~$ sudo tcpdump -i lo tcp -c 10 -w ~/syn_test.pcap & sleep 1 && sudo nmap -sS 127.0.0.1 -p 80
[1] 25104
tcpdump: listening on lo, link-type EN10MB (Ethernet), snapshot length 262144 bytes
sudo: 'nmap': command not found
it-24@IT-24:~$ sudo suricata -c /etc/suricata/suricata.yaml -k none -l /tmp/suricata_test -r ~/syn_test.pcap
i: suricata: This is Suricata version 8.0.3 RELEASE running in USER mode
i: mpm-hs: Rule group caching - loaded: 0 newly cached: 0 total cacheable: 0
E: pcap: truncated dump file; tried to read 4 file header bytes, only got 0
W: pcap: Failed to init pcap file /home/it-13/syn_test.pcap, skipping
E: pcap: pcap file reader thread failed to initialize
i: threads: Threads created -> RX: 1 W: 12 FM: 1 FR: 1 Engine started.
i: suricata: Signal Received. Stopping engine.
it-24@IT-24:~$ sudo apt update && sudo apt install nmap -y
Hit:1 http://security.ubuntu.com/ubuntu resolute-security InRelease
Hit:2 http://archive.ubuntu.com/ubuntu resolute InRelease
Hit:3 http://archive.ubuntu.com/ubuntu resolute-updates InRelease
Hit:4 http://archive.ubuntu.com/ubuntu resolute-backports InRelease
2 packages can be upgraded. Run 'apt list --upgradable' to see them.
Installing:
nmap

Installing dependencies:
libblas3 liblinear4 liblua5.4-0 nmap-common

Suggested packages:
liblinear-tools liblinear-dev ncat ndiff zenmap

Summary:
Upgrading: 0, Installing: 5, Removing: 0, Not Upgrading: 2
Download size: 6688 kB
Space needed: 29.0 MB / 1022 GB available

Get:1 http://archive.ubuntu.com/ubuntu resolute/main amd64 libblas3 amd64 3.12.1-7ubuntu1 [260 kB]
Get:2 http://archive.ubuntu.com/ubuntu resolute/universe amd64 liblinear4 amd64 2.3.0+dfsg-5build2 [44.8 kB]
Get:3 http://archive.ubuntu.com/ubuntu resolute/main amd64 liblua5.4-0 amd64 5.4.8-1build1 [163 kB]
Get:4 http://archive.ubuntu.com/ubuntu resolute/universe amd64 nmap-common all 7.98+dfsg-1 [4517 kB]
Get:5 http://archive.ubuntu.com/ubuntu resolute/universe amd64 nmap amd64 7.98+dfsg-1 [1704 kB]
Fetched 6688 kB in 1s (6590 kB/s)
Selecting previously unselected package libblas3:amd64.

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it-24@IT24:~$
Processing triggers for man-db (2.13.1-1build1) ...
Processing triggers for libc-bin (2.43-2ubuntu2.3) ...
it-24@IT24:~$ sudo tcpdump -i lo tcp -c 10 -w ~/syn_test.pcap & sleep 1 && sudo nmap -sS 127.0.0.1 -p 80
[2] 25391
tcpdump: listening on lo, link-type EN10MB (Ethernet), snapshot length 262144 bytes
Starting Nmap 7.98 ( https://nmap.org ) at 2026-08-24 04:53 +0000
Nmap scan report for localhost (127.0.0.1)
Host is up (0.00042s latency).

PORT      STATE SERVICE
80/tcp    closed http

Nmap done: 1 IP address (1 host up) scanned in 0.07 seconds
it-24@IT24:~$ sudo suricata -c /etc/suricata/suricata.yaml -k none -l /tmp/suricata_test -r ~/syn_test.pcap
i: suricata: This is Suricata version 8.0.3 RELEASE running in USER mode
i: mpm-hs: Rule group caching - loaded: 0 newly cached: 0 total cacheable: 0
E: pcap: truncated dump file; tried to read 4 file header bytes, only got 0
W: pcap: Failed to init pcap file /home/it-13/syn_test.pcap, skipping
E: pcap: pcap file reader thread failed to initialize
i: threads: Threads created -> RX: 1 W: 12 FM: 1 FR: 1 Engine started.
i: suricata: Signal Received. Stopping engine.
it-24@IT24:~$ sudo cat /tmp/suricata_test/fast.log
08/24/2026-04:51:03.625457 [**] [1:1000001:1] ICMP Ping Detected [**] [Classification: (null)] [Priority: 3] {ICMP} 127.0.0.1:8 -> 127.0.0.1:0
it-24@IT24:~$ sudo tcpdump -i lo tcp -w ~/syn_test.pcap &
TCP_PID=$!
sleep 1
sudo nmap -sS 127.0.0.1 -p 80
sudo kill $TCP_PID
[3] 25455
tcpdump: listening on lo, link-type EN10MB (Ethernet), snapshot length 262144 bytes
Starting Nmap 7.98 ( https://nmap.org ) at 2026-08-24 04:54 +0000
Nmap scan report for localhost (127.0.0.1)
Host is up (0.00016s latency).

PORT      STATE SERVICE
80/tcp    closed http

Nmap done: 1 IP address (1 host up) scanned in 0.08 seconds
0 packets captured
4 packets received by filter
0 packets dropped by kernel

it-24@IT24:~$
Starting Nmap 7.98 ( https://nmap.org ) at 2026-08-24 04:54 +0000
Nmap scan report for localhost (127.0.0.1)
Host is up (0.00016s latency).

PORT      STATE SERVICE
80/tcp    closed http

Nmap done: 1 IP address (1 host up) scanned in 0.08 seconds
0 packets captured
4 packets received by filter
0 packets dropped by kernel

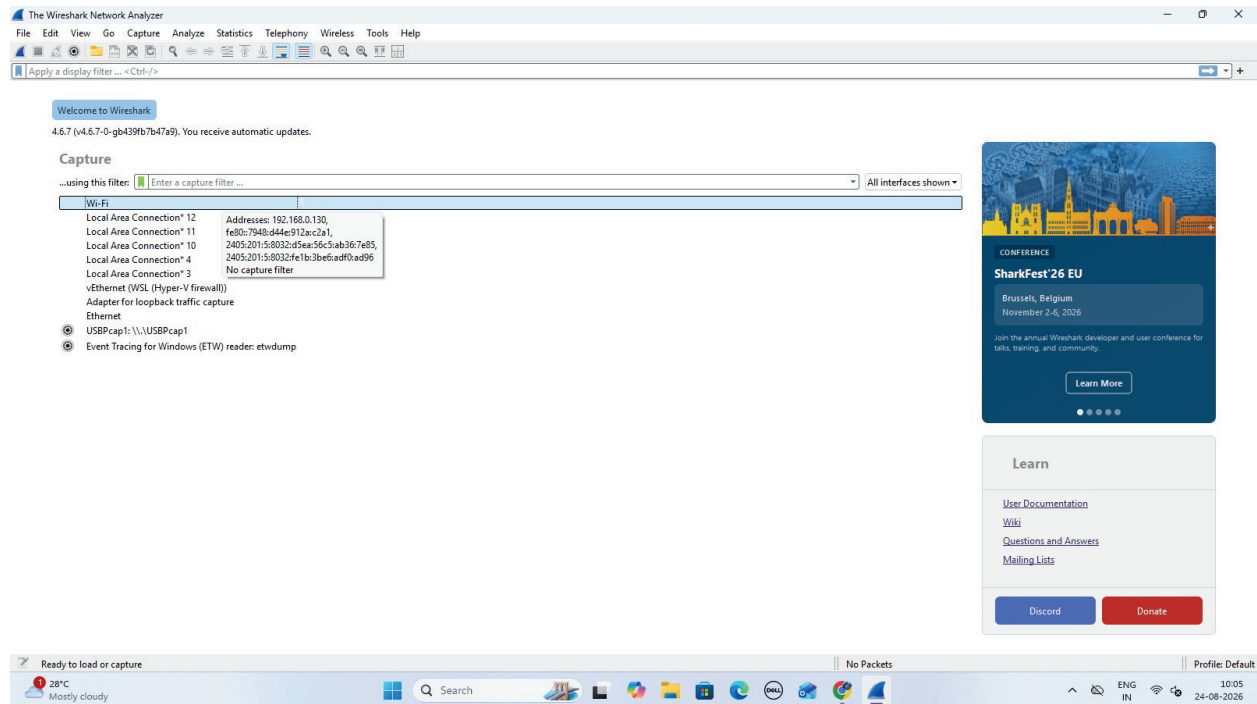
-r ~/syn_test.pcap
sudo cat /tmp/suricata_test/fast.log
i: suricata: This is Suricata version 8.0.3 RELEASE running in USER mode
i: mpm-hs: Rule group caching - loaded: 0 newly cached: 0 total cacheable: 0
E: pcap: failed to get first packet timestamp. pcap_next_ex(): -2
W: pcap: Failed to init pcap file /home/it-13/syn_test.pcap, skipping
E: pcap: pcap file reader thread failed to initialize
i: threads: Threads created -> RX: 1 W: 12 FM: 1 FR: 1 Engine started.
i: suricata: Signal Received. Stopping engine.
[3]+ Done sudo tcpdump -i lo tcp -w ~/syn_test.pcap
08/24/2026-04:51:03.625457 [**] [1:1000001:1] ICMP Ping Detected [**] [Classification: (null)] [Priority: 3] {ICMP} 127.0.0.1:8 -> 127.0.0.1:0
it-24@IT24:~$ sudo timeout 3s tcpdump -i lo tcp -w ~/syn_test.pcap >/dev/null 2>&1 & sleep 1 && sudo nmap -sS 127.0.0.1 -p 80 >/dev/null && sleep 3 && sudo suricata -c /etc/suricata/suricata.yaml -k none -l /tmp/suricata_test -r ~/syn_test.pcap && sudo cat /tmp/suricata_test/fast.log
[3] 25537
i: suricata: This is Suricata version 8.0.3 RELEASE running in USER mode
i: mpm-hs: Rule group caching - loaded: 0 newly cached: 0 total cacheable: 0
i: threads: Threads created -> RX: 1 W: 12 FM: 1 FR: 1 Engine started.
i: suricata: Signal Received. Stopping engine.
i: pcap: read 1 file, 2 packets, 112 bytes
08/24/2026-04:51:03.625457 [**] [1:1000001:1] ICMP Ping Detected [**] [Classification: (null)] [Priority: 3] {ICMP} 127.0.0.1:8 -> 127.0.0.1:0
08/24/2026-04:54:48.131868 [**] [1:1000002:1] Possible TCP SYN Scan Detected [**] [Classification: (null)] [Priority: 3] {TCP} 127.0.0.1:60252 -> 127.0.0.1:80
[3]+ Exit 120 sudo timeout 3s tcpdump -i lo tcp -w ~/syn_test.pcap >/dev/null 2>&1
it-24@IT24:~$ ^C
it-24@IT24:~$ ^C
it-24@IT24:~$ rm -f ~/test.pcap ~/icmp_test.pcap ~/syn_test.pcap
it-24@IT24:~$

```

OUTPUT: WIRESHARK

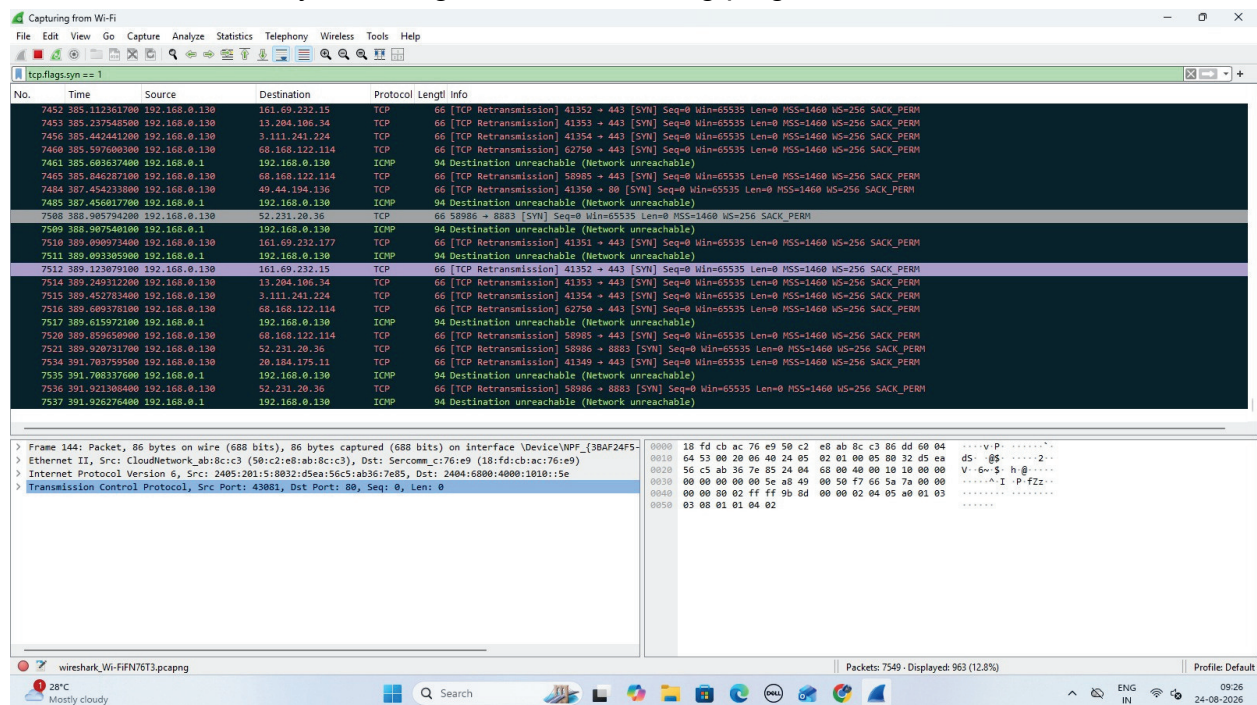
STEPS :

1 . Open Wireshark and select your active network interface.



2 . Start capturing network packets.

3 . Generate traffic by browsing a website or using ping



- 4 . Apply filters like **tcp**, **udp**, **icmp**, or **http**.
- 5 . Analyze packets for suspicious or unusual traffic.
- 6 . Check source IP, destination IP, ports, and protocols.
- 7 . Identify potential malicious activity or security breaches.

● **icmp** :

The screenshot shows a Wireshark capture of ICMP traffic. The packet list pane displays several ICMP Destination Unreachable (Type 3) packets from 192.168.0.1 to 192.168.0.130. The packet details pane for the selected packet shows the ICMP header and the Internet Protocol Version 4 header. The packet bytes pane shows the raw data of the packet.

No.	Time	Source	Destination	Protocol	Length	Info
10006	50.023654000	192.168.0.1	192.168.0.130	ICMP	94	Destination unreachable (Network unreachable)
10051	510.434324100	192.168.0.1	192.168.0.130	ICMP	94	Destination unreachable (Network unreachable)
10052	510.434325000	192.168.0.1	192.168.0.130	ICMP	94	Destination unreachable (Network unreachable)
10053	510.434325400	192.168.0.1	192.168.0.130	ICMP	94	Destination unreachable (Network unreachable)
10055	510.449237700	192.168.0.1	192.168.0.130	ICMP	94	Destination unreachable (Network unreachable)
10096	511.326668900	192.168.0.1	192.168.0.130	ICMP	94	Destination unreachable (Network unreachable)

Packet details for the selected packet (No. 10096):

- Frame 133: Packet, 94 bytes on wire (752 bits), 94 bytes captured (752 bits) on interface \Device\NPF... [30AF24F1]
- Ethernet II, Src: TpLinkTechno_9c:44:d4 (60:32:b1:9c:44:d4), Dst: Cloudnetwor_ab8c:c3 (50:c2:e8:ab:8c:c3)
- Internet Protocol Version 4, Src: 192.168.0.1, Dst: 192.168.0.130
- Internet Control Message Protocol (ICMP)
 - Type: Destination unreachable (3)
 - Code: 0 (Network unreachable)
 - Checksum: 0x9c4b [correct]
 - [Checksum Status: Good]
 - Unused: 00000000
 - Internet Protocol Version 4, Src: 192.168.0.130, Dst: 20.50.201.200
 - 0100 = Version: 4
 - 0101 = Header Length: 20 bytes (5)
 - Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
 - 0000 00.. = Differentiated Services Codepoint: Default (0)
 -00 = Explicit Congestion Notification: Not ECN-Capable Transport (0)
 - Total Length: 52
 - Identification: 0x0c32 (3122)
 - 010. = Flags: 0x2, Don't fragment
 - ... 0 0000 0000 = Fragment Offset: 0
 - Time to Live: 128
 - Protocol: TCP (6)
 - Header Checksum: 0x4fd6 [validation disabled]
 - [Header checksum status: Unverified]
 - Source Address: 192.168.0.130
 - Destination Address: 20.50.201.200
 - [Stream index: 36]
 - Transmission Control Protocol, Src Port: 22471, Dst Port: 443, Seq: 1810044553
 - Source Port: 22471
 - Destination Port: 443
 - Sequence Number: 1810044553
 - [Stream index: 3]
 - [Stream Packet Number: 2]

Summary: Internet Control Message Protocol (icmp), 60 bytes

Packets: 10145 - Displayed: 489 (4.8%)

● **Tcp.analysis.retransmission:**

The screenshot shows a Wireshark capture of TCP traffic. The packet list pane displays several TCP retransmission packets from 192.168.0.1 to 192.168.0.130. The packet details pane for the selected packet shows the TCP header and the Internet Protocol Version 4 header. The packet bytes pane shows the raw data of the packet.

No.	Time	Source	Destination	Protocol	Length	Info
14688	653.487447500	192.168.0.130	52.168.112.66	TCP	66	[TCP Retransmission] 10531 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
14689	653.492789600	192.168.0.1	192.168.0.130	ICMP	94	Destination unreachable (Network unreachable)
14718	654.769580800	192.168.0.130	20.184.175.11	TCP	66	[TCP Retransmission] 10532 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
14719	654.773131900	192.168.0.1	192.168.0.130	ICMP	94	Destination unreachable (Network unreachable)
14733	655.429311000	192.168.0.130	161.69.237.219	TCP	66	[TCP Retransmission] 10533 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
14732	655.598275800	192.168.0.1	192.168.0.130	ICMP	94	Destination unreachable (Network unreachable)
14756	656.610598500	192.168.0.130	161.69.237.219	TCP	66	[TCP Retransmission] 10533 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
14757	656.614587600	192.168.0.1	192.168.0.130	ICMP	94	Destination unreachable (Network unreachable)
14789	658.621936900	192.168.0.130	161.69.237.219	TCP	66	[TCP Retransmission] 10533 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
14790	658.624114300	192.168.0.1	192.168.0.130	ICMP	94	Destination unreachable (Network unreachable)
14798	658.778424000	192.168.0.130	20.184.175.11	TCP	66	[TCP Retransmission] 10532 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
14799	658.778762400	192.168.0.1	192.168.0.130	ICMP	94	Destination unreachable (Network unreachable)

Packet details for the selected packet (No. 14799):

- Source Port: 7239
- Destination Port: 443
- [Stream index: 5]
- [Stream Packet Number: 3]
- [Conversation completeness: Incomplete, SYN_SENT (1)]
- [TCP Segment Len: 0]
- Sequence Number: 0 (relative sequence number)
- Sequence Number (raw): 1861242804
- [Next Sequence Number: 1 (relative sequence number)]
- Acknowledgment Number: 0
- Acknowledgment number (raw): 0
- 1000 = Header Length: 32 bytes (8)
- Flags: 0x002 (SYN)
- Window: 65535
- [Calculated window size: 65535]
- Checksum: 0xd4d3 [unverified]
- [Checksum Status: Unverified]
- Urgent Pointer: 0
- Options: (12 bytes), Maximum segment size, No-Operation (NOP), Window scale, No-Operation (NOP), No-Operation (NOP)
 - TCP Option - Maximum segment size: 1460 bytes
 - TCP Option - No-Operation (NOP)
 - TCP Option - Window scale: 8 (multiply by 256)
 - TCP Option - No-Operation (NOP)
 - TCP Option - No-Operation (NOP)
 - TCP Option - No-Operation (NOP)
 - TCP Option - SACK permitted
 - [Timestamps]
 - [SEQ/ACK analysis]

Summary: wireshark-Wi-Fi\7673.pcapng

Packets: 14834 - Displayed: 1576 (10.6%)

- tcp.flags.syn == 1 && tcp.flags.ack == 0 :

Frame 191: Packet, 66 bytes on wire (528 bits), 66 bytes captured (528 bits) on interface \Device\NPF_{3BAF24F1...}

Ethernet II, Src: CloudNetwork_ab8c3c3 (50:c2:e8:ab:8c:3c3), Dst: TplinkTechno_Bb:ff:67 (74:da:88:b8:ff:67)

Internet Protocol Version 4, Src: 192.168.0.130, Dst: 140.82.112.25

TCP, Src Port: 7239, Destination Port: 443, Seq: 0, Len: 0

Transmission Control Protocol, Src Port: 7239, Dst Port: 443, Seq: 0, Len: 0

This frame is a (suspected) retransmission: Label

Packets: 16589 - Displayed: 866 (5.2%)

- Http:

Frame 147: Packet, 276 bytes on wire (2208 bits), 276 bytes captured (2208 bits) on interface \Device\NPF_{3BAF24F1...}

Ethernet II, Src: CloudNetwork_ab8c3c3 (50:c2:e8:ab:8c:3c3), Dst: Sercomm_c:76:e9 (18:fd:cb:ac:76:e9)

Internet Protocol Version 4, Src: 2405:201:5:8032:d5e::, Dst: 2404:6800:4000:1010::

Transmission Control Protocol, Src Port: 43081, Dst Port: 80, Seq: 1, Ack: 1, Len: 202

Hypertext Transfer Protocol

GET /r/gsr1.cr1 HTTP/1.1

Host: 192.168.0.130

User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/88.0.4399.72 Safari/537.36

Accept: */*

Accept-Encoding: gzip, deflate, br

Accept-Language: en-US,en;q=0.9

Cache-Control: no-cache

Connection: keep-alive

Cookie: ...

Tls :

Transport Layer Security: Protocol

Packets: 18089 - Displayed: 899 (5.0%)

Profile: Default

8 . Stop the capture and save the packet file for analysis.

9 . Record the detected suspicious packets as IDS observations.

Wi-Fi <live capture in progress>

Packets: 381

Profile: Default